to Einstein, was “the complete description of any (individual) real situation (as it supposedly exists irrespective of any act of observation or substantiation).” In this view, Einstein knew he was out of sync with the intellectual fashion of his day: “Whenever the positivistically inclined modern physicist hears such a formulation his reaction is that of a pitying smile.” But Einstein found positivism wholly uncompelling, seeing it as a total rejection of the idea of a physical world, tantamount to claiming that reality exists only in our minds: “What I dislike in this kind of argumentation is the basic positivistic attitude, which from my point of view is untenable, and which seems to me to come to the same thing as [Irish philosopher George] Berkeley’s principle, *esse est percipi* [to be is to be perceived].” Though Einstein had no doubts about the importance of the new quantum theory, he was convinced that Born and Heisenberg were wrong to claim that quantum physics was complete, and that Bohr’s philosophy of complementarity was inadequate for understanding the true nature of the quantum world. His thought experiment was simple, elegant, and carefully designed to strike at the heart of this inadequacy.

Consider, Einstein said to the assembled Solvay conference, a stream of electrons passing through a very small hole in a screen (Figure 3.2). On the other side of the screen, there is a hemisphere of phosphorescent film that can register the impact of a single electron. According to quantum physics, the wave function for the stream of electrons should be uniform—the probability of an electron hitting the film should be the same at every location on the hemisphere. And that’s fine—if quantum physics says you’ll find ten electrons per square centimeter of film after conducting your experiment, then on average, that’s what you’ll find. Quantum physics is great at describing the aggregate behavior of large groups of particles. But quantum physics can’t do more than assign probabilities; it can’t tell you exactly how many electrons will hit each part of the screen, it can only give you an average.